

Agisoft Metashape

Processing Report
28 February 2026



Survey Data

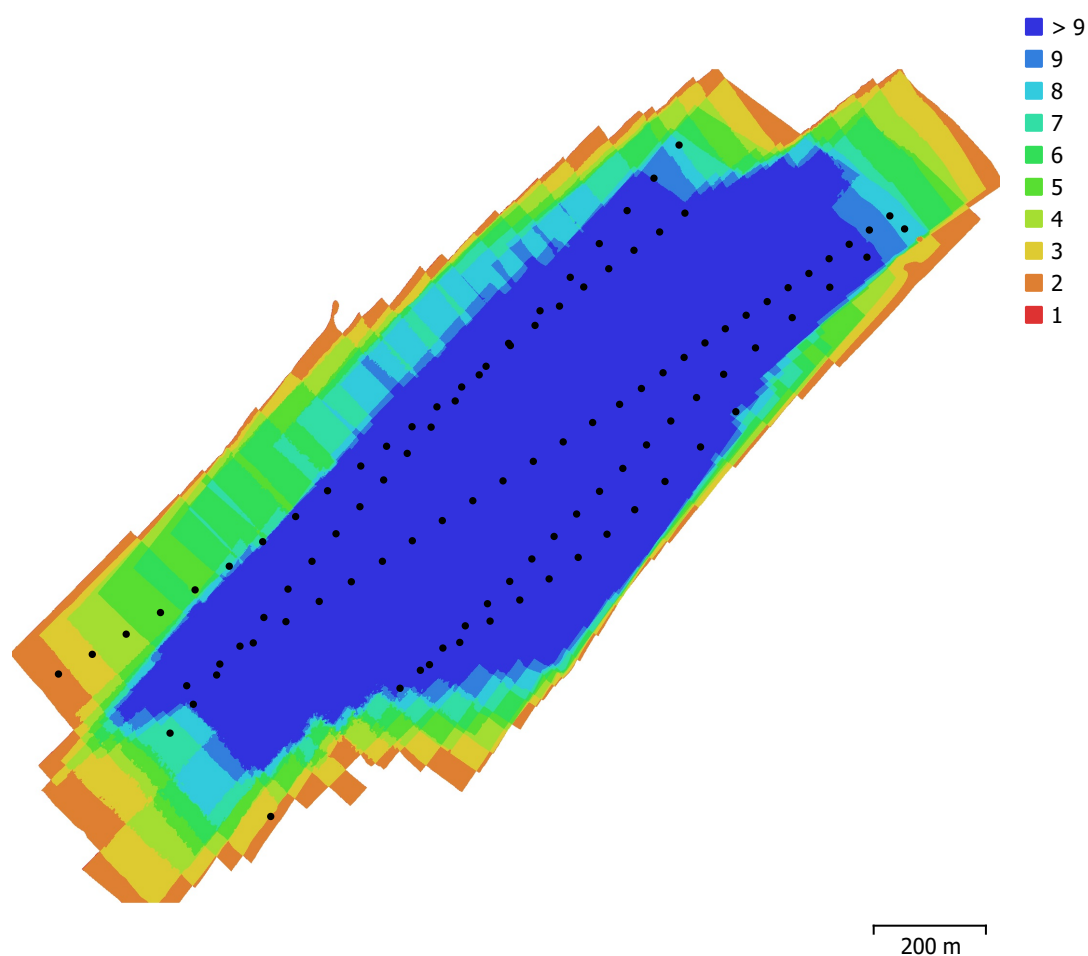


Fig. 1. Camera locations and image overlap.

Number of images:	104	Camera stations:	104
Flying altitude:	427 m	Tie points:	86,336
Ground resolution:	9.96 cm/pix	Projections:	394,464
Coverage area:	1.15 km ²	Reprojection error:	0.658 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
NIKON D3 (35mm)	4256 x 2832	35 mm	8.46 x 8.46 μm	No

Table 1. Cameras.

Camera Calibration

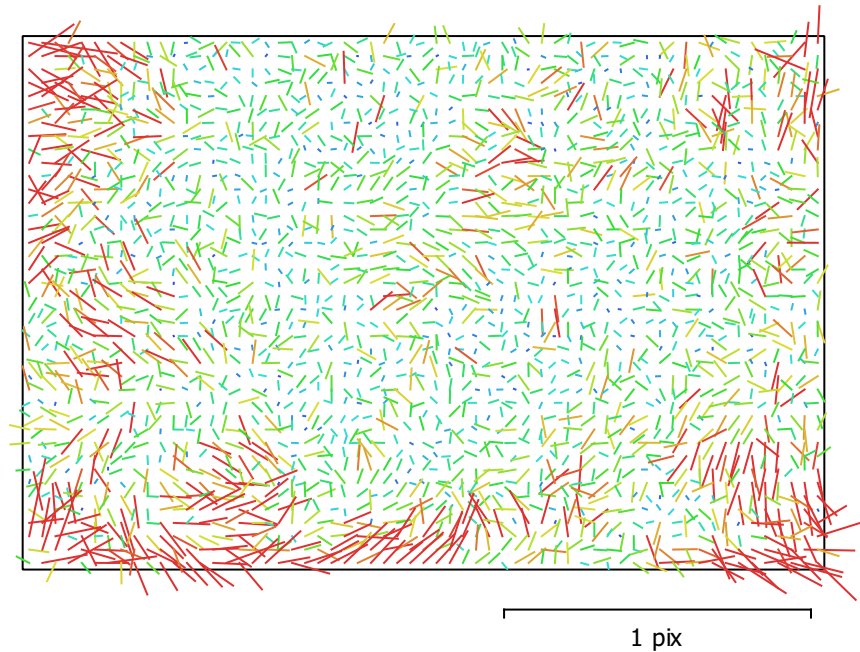


Fig. 2. Image residuals for NIKON D3 (35mm).

NIKON D3 (35mm)

104 images

Type	Resolution	Focal Length	Pixel Size
Frame	4256 x 2832	35 mm	8.46 x 8.46 μm

	Value	Error	F	Cx	Cy	K1	K2	K3	P1	P2
F	4255.76	0.12	1.00	0.40	-0.15	-0.17	0.13	-0.10	-0.11	0.04
Cx	15.4716	0.054		1.00	0.01	-0.04	0.02	-0.01	0.71	0.11
Cy	-16.5747	0.035			1.00	0.04	-0.03	0.03	0.13	0.38
K1	-0.0951215	6.7e-05				1.00	-0.97	0.92	0.01	-0.06
K2	0.125494	0.00045					1.00	-0.98	0.00	0.05
K3	-0.0388532	0.00091						1.00	-0.00	-0.04
P1	-0.000413234	3.3e-06							1.00	0.09
P2	-0.000342327	1.8e-06								1.00

Table 2. Calibration coefficients and correlation matrix.

Camera Locations

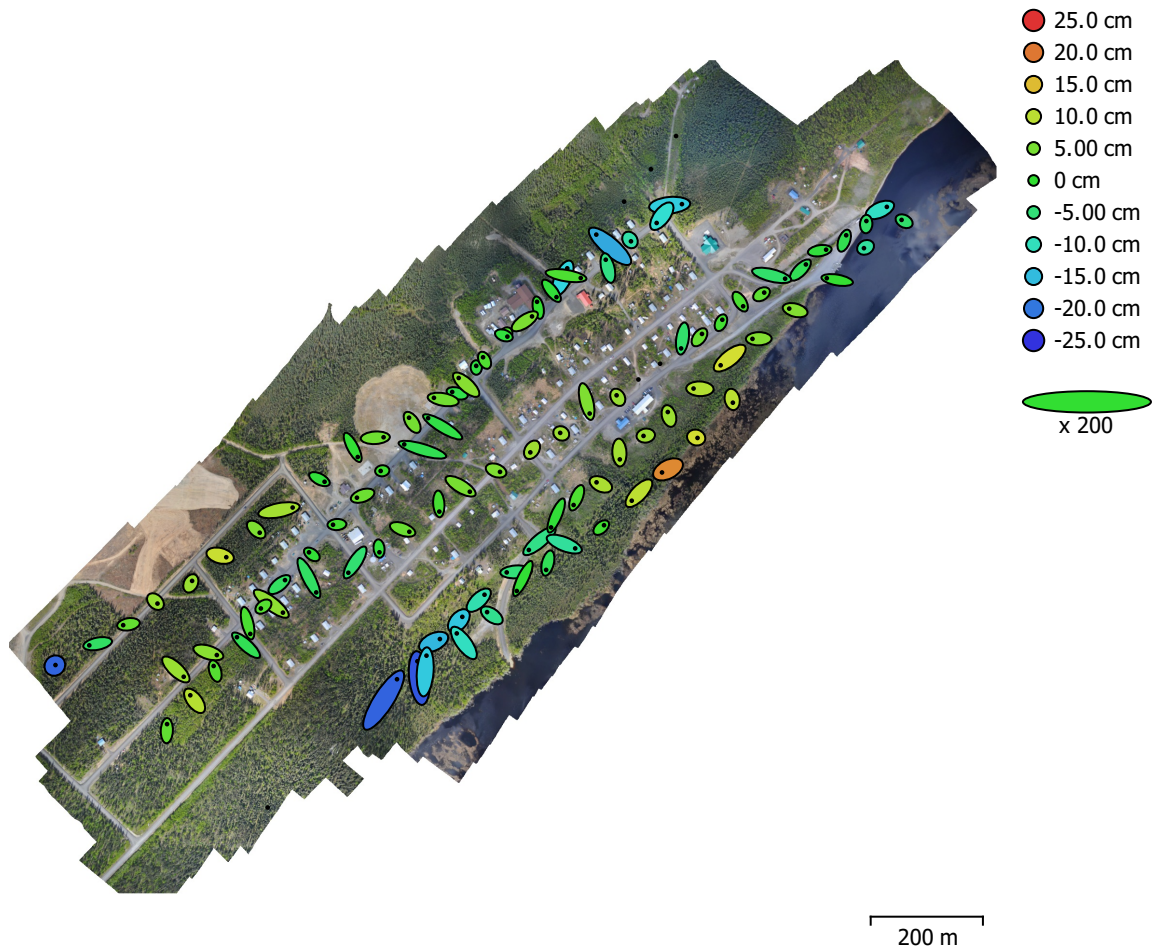


Fig. 3. Camera locations and error estimates.

Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.

Estimated camera locations are marked with a black dot.

X error (cm)	Y error (cm)	Z error (cm)	XY error (cm)	Total error (cm)
11.5461	11.2096	7.77999	16.0925	17.8745

Table 3. Average camera location error.

X - Easting, Y - Northing, Z - Altitude.

Ground Control Points

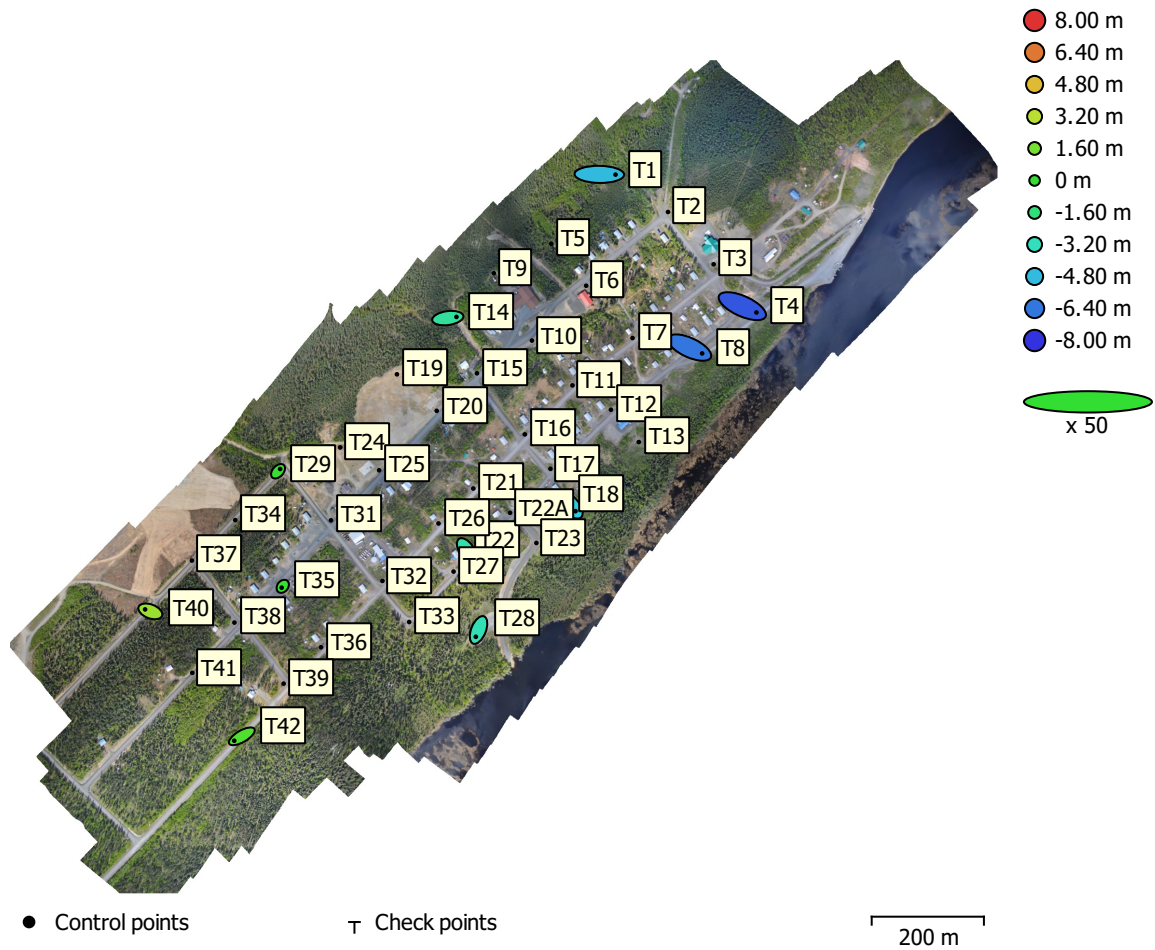


Fig. 4. GCP locations and error estimates.

Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.

Estimated GCP locations are marked with a dot or crossing.

Count	X error (m)	Y error (m)	Z error (m)	XY error (m)	Total (m)
11	0.620489	0.312327	3.87651	0.694661	3.93826

Table 4. Control points RMSE.

X - Easting, Y - Northing, Z - Altitude.

Label	X error (m)	Y error (m)	Z error (m)	Total (m)	Image (pix)
T1	1.1553	-0.0168976	-4.84082	4.9768	0.403 (7)
T2					
T3					
T4	1.00507	-0.452136	-7.15213	7.23654	0.317 (16)
T5					
T6					
T7					
T8	0.903418	-0.416874	-6.41911	6.49576	0.211 (17)
T9					
T10					
T11					
T12					
T13					
T14	0.618494	0.0843941	-2.42844	2.50739	0.200 (9)
T15					
T16					
T17					
T18	0.37095	-0.422282	-4.50399	4.53893	0.227 (20)
T19					
T20					
T21					
T22	0.233767	-0.34564	-2.80273	2.83362	0.287 (27)
T22A					
T23					
T24					
T25					
T26					
T27					
T28	-0.195641	-0.484621	-3.00516	3.05026	0.298 (13)
T29	0.148949	0.161488	0.153704	0.268121	0.299 (6)
T31					

Label	X error (m)	Y error (m)	Z error (m)	Total (m)	Image (pix)
T32					
T33					
T34					
T35	-0.077764	-0.0975099	0.160055	0.202912	0.274 (19)
T36					
T37					
T38					
T39					
T40	-0.371794	0.141084	2.42928	2.46162	0.264 (5)
T41					
T42	-0.542671	-0.315072	0.699189	0.939483	0.230 (11)
Total	0.620489	0.312327	3.87651	3.93826	0.273

Table 5. Control points.
X - Easting, Y - Northing, Z - Altitude.

Digital Elevation Model

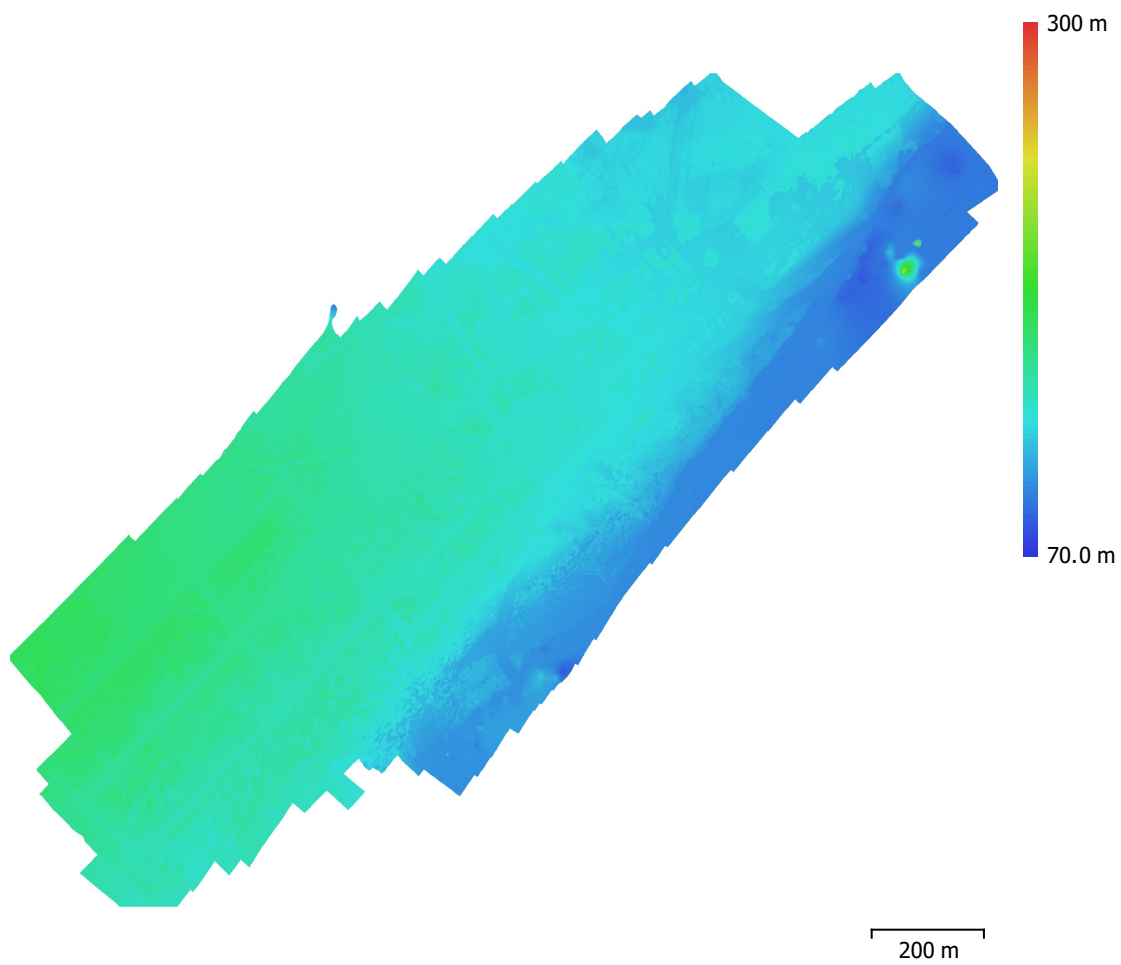


Fig. 5. Reconstructed digital elevation model.

Resolution: 19.9 cm/pix
Point density: 25.2 points/m²

Processing Parameters

General

Images	104
Aligned images	104
Markers	42
Coordinate system	WGS 84 / UTM zone 6N (EPSG::32606)
Rotation angles	Yaw, Pitch, Roll

Tie Points

Points	86,336 of 92,724
RMS reprojection error	0.212409 (0.657758 pix)
Max reprojection error	0.833449 (22.6315 pix)
Mean key point size	2.72393 pix
Point colors	3 bands, uint8
Key points	No
Average tie point multiplicity	4.85861

Alignment parameters

Accuracy	Highest
Generic preselection	Yes
Reference preselection	Source
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	4,000
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	No
Matching time	1 minutes 1 seconds
Matching memory usage	383.81 MB
Alignment time	14 seconds
Alignment memory usage	55.42 MB

Optimization parameters

Parameters	f, cx, cy, k1-k3, p1, p2
Adaptive camera model fitting	No
Exclude corners	No
Optimization time	1 seconds
Date created	2026:02:28 13:28:21
Software version	2.3.0.21954
File size	9.13 MB

Depth Maps

Count	104
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Depth maps generation parameters

Quality	High
Filtering mode	Mild
Max neighbors	16
Processing time	1 minutes 48 seconds
Memory usage	2.07 GB
Date created	2026:02:28 15:14:06
Software version	2.3.0.21954
File size	507.42 MB

Point Cloud

Points	45,088,549
Coordinate precision	4.98 cm

Point attributes	
Color	3 bands, uint8
Normal	
Point classes	
Created (never classified)	45,088,549
Depth maps generation parameters	
Quality	High
Filtering mode	Mild
Max neighbors	16
Processing time	1 minutes 48 seconds
Memory usage	2.07 GB
Point cloud generation parameters	
Source data	Depth maps
Processing time	4 minutes 2 seconds
Memory usage	3.28 GB
Date created	2026:02:28 15:18:10
Software version	2.3.0.21954
File size	587.05 MB
Model	
Faces	8,907,816
Vertices	4,458,719
Vertex colors	3 bands, uint8
Depth maps generation parameters	
Quality	High
Filtering mode	Mild
Max neighbors	16
Processing time	1 minutes 48 seconds
Memory usage	2.07 GB
Point cloud generation parameters	
Source data	Depth maps
Processing time	4 minutes 2 seconds
Memory usage	3.28 GB
Reconstruction parameters	
Surface type	Height field
Source data	Point cloud
Interpolation	Enabled
Strict volumetric masks	No
Processing time	2 minutes 13 seconds
Memory usage	1.37 GB
Date created	2026:02:28 15:19:17
Software version	2.3.0.21954
File size	174.22 MB
DEM	
Size	8,839 x 7,445
Resolution	19.9 cm/pix
Coordinate system	WGS 84 / UTM zone 6N (EPSG::32606)
Reconstruction parameters	
Source data	Point cloud
Interpolation	Enabled
Processing time	22 seconds
Memory usage	310.57 MB
Date created	2026:02:28 15:23:48
Software version	2.3.0.21954
File size	129.62 MB
Orthomosaic	
Size	17,672 x 14,902

Resolution	9.95 cm/pix
Coordinate system	WGS 84 / UTM zone 6N (EPSG::32606)
Colors	3 bands, uint8
Orthophotos	490.65 MB

Reconstruction parameters

Blending mode	Mosaic
Surface	DEM
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	1 minutes 17 seconds
Memory usage	1.45 GB
Date created	2026:02:28 14:07:25
Software version	2.3.0.21954
File size	623.31 MB

System

Software name	Agisoft Metashape Professional
Software version	2.3.0 build 21954
OS	Windows 64 bit
RAM	63.92 GB
CPU	AMD Ryzen 7 5800XT 8-Core Processor
GPU(s)	NVIDIA GeForce RTX 5070 Ti